# Ultra-sensitive Distributed Acoustic Sensor using "*Supercoherent*" Continuous-wave Photons

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***Abstract*—Distributed sensing of acoustic or seismic waves plays a crucial role in life-saving disaster management, strategic applications, industrial monitoring, scientific exploration, and structural health diagnostics. Optical fiber–based distributed sensing platforms offer a unique combination of long-range coverage, intrinsic multiplexing, and remote interrogation, enabling a single interrogator to monitor multiple sensing channels with minimal cabling. This scalability translates into a cost-effective, robust, and highly deployable monitoring architecture. In contrast to conventional pulsed-laser schemes that infer perturbations from weak backscattered signals in the sensing fiber, we introduce a fundamentally different approach based on a continuous-wave, nearly monochromatic probe laser exhibiting ultra-low linear frequency drift. This "*Supercoherent*" laser–based Distributed Acoustic Sensing (SL-DAS) achieves unprecedented sensing accuracy across an exceptionally broad bandwidth spanning sub-mHz to kHz frequencies. The approach enables substantially extended sensing lengths, enhanced spatial localization, and markedly improved frequency resolution compared to traditional techniques. By employing dual self-heterodyne detection to directly transduce perturbation-induced phase noise in the probe laser, SL-DAS becomes intrinsically immune to system drift. Beyond significantly advancing existing DAS capabilities, this platform opens new frontiers with transformative implications for distributed quantum sensing [1], [2], gravitational-wave detector bandwidth extension, dark-matter searches [3], biomedical diagnostics [4], and soft-landing technologies.**

## I. Introduction

Fibre-optic sensors have attracted significant attention due to their wide-area distributed sensing capability along with numerous other advantages, such as robustness, remote operation, and high sensitivity [5]. Unlike static measurement techniques, distributed sensors also record fast temporal variations, thereby providing more comprehensive and dynamic information. In comparison to traditional sensors, fibre-optic ones are lightweight, compact, power-efficient, have a wide operational bandwidth, and offer fast response due to the use of photonic elements. They are also immune to any electromagnetic interference, which makes them spoofing-free and capable of functioning reliably in faraway and harsh environments without any *in-situ* human intervention [6], [7]. Due to these advantages, optical fiber based temperature [8]–[15], pressure [16], stress and deformations [17]–[19], rotation [20], electrical discharge [21], acoustic [22]–[27], seismic [28], [29], and various other sensors have gained enormous demands.

Fibre-optic systems offer multiplexing capability, integrating multiple sensing fibers into one interrogator to operate using a common light source, optical components, and detection systems. This not only makes the networked sensor cost-effective but also power-efficient, reducing system size and minimizing differential systematics among all the channels. Recent advancements have enabled such systems to sense at every point along a single fiber and spatially resolve with an accuracy of a few meters, thereby replacing numerous discrete sensors, simplifying and scaling up the sensing network without compromising the accuracy and reliability [30]. Easy to use distributed fibre-optic sensors have been rapidly adopted across a wide range of research fields: particularly to sense the ground vibrations affecting large-scale detectors and networked experiments, e.g., particle accelerators, storage rings, radio-antenna arrays, geophysical studies [31], [32], vertical seismic profiling [33]–[35], microseismic studies [36], [37], gravitational wave detection [38], [39]; strategic applications: e.g., monitoring of hydraulic fracturing processes [40]–[42], perimeter surveillance [43]–[45], satellite free navigation; as well as industrial uses: e.g., health monitoring of flyover, railway tracks [46], and runways [47], early warnings [29], [48], and many more. Their key strength lies in their ability to sense multiple parameters, spatially resolved continuous measurements over a large area, with a wide frequency bandwidth and high sensitivity. The reported optical fiber based distributed sensing relies on Rayleigh, Brillouin, and Raman scatterings of photons occurring from a perturbed fiber [5], [30], [49], where amplitude, frequency, and polarization of the scattered photons are influenced by changes in physical parameters, thereby enabling their measurement; at the same time, they are indistinguishable from unwanted scatterings arising from manufacturing defects intrinsic to the fibers.

Here, we report a "*Supercoherent*" Laser-based Distributed Acoustic Sensing (SL-DAS) that utilizes a nearly monochromatic, ultra-low drift continuous-wave probe laser for the sensing and dual-self heterodyne for the detections, which enhances the sensitivity compared to pulsed laser-induced scattering phenomena-based DAS. Using normal telecom fibers SL-DAS is cabale to detect environmental deformations and strains with sub-m localization capabilities over a sub-mHz to kHz bandwidth, thereby significantly leveraging capabilities in the currently used cases and potentially opening up new frontiers of applications.

## II. Super Coherent Laser Based Distributed Acoustic Sensing

The SL-DAS setup is an end-to-end optical fibre-coupled system that runs from the lab (L) to a remote (R) site using a phase-coherent fibre (PCF) [ref. to Stanley paper] and a

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co-propagating, unstabilized fibre for sensing. The fiber links, along with the optical arrangement, are shown in Fig. 1(a). A lab-generated continuous wave (CW) nearly monochromatic $\sim 1\,\mathrm{Hz}$ linewidth at $1\,\mathrm{s}$ and $33\,\mathrm{mHz/s}$ linear drifted, therefore ultrastable, laser at $1550\,\mathrm{nm}$ wavelength, namely the super-coherent laser (SL) is transported to R ($90\,\%$ power) via the PCF link where it is coupled to the sensing fibre, whereas, the remaining $10\,\%$ is used as one optical frequency reference (OFR1). The $90\,\%$ is further split into $9:1$ ratio using a coupler (Cpl1), of which $90\,\%$ passes through an acousto optic modulator (AOM1) and $10\,\%$ is reflected by the Faraday mirror (FM1) and is used as the second reference OFR2. After the AOM1, light propagates to R through the PCF where it is equally split into two by Cpl2; one portion of it is reflected by FM2 and returned to the lab via the PCF that is used for stabilizing this channel [ref. to Stanley paper] and the other portion propagates to L through the sensing fibre upon frequency modulation by AOM2. After AOM2, the light is again split into $9:1$ ratio, of which $90\,\%$ is reflected by FM 3 and propagates back through AOM2, then sensing $\rightarrow$ PCF followed by AOM 1, and returns to L. The remaining $10\,\%$ interferes with OFR 1 at Cpl. 4, and the resultant beat frequency is measured using a fast photodiode (PD2), whereas the OFR2 and the double-passed light interfere at Cpl1, and the beat frequency is detected at PD1. Both the AOMs operate at $20\,\mathrm{MHz}$ and their $-1$-order diffracted lights are used in the experiment. Optical isolators (ISO) at $-45\,\mathrm{dB}$ are used prior to connecting the beat signals to PD 1 and 2, for adequate suppression of the back-reflected photons. The PD1 and PD2, therefore, detects $\delta\nu_{\mathrm{in}}(t) = 80\,\mathrm{MHz} - \chi_{\mathrm{in}}(t)$ and $\delta\nu_{\mathrm{out}}(t) = 40\,\mathrm{MHz} - \chi_{\mathrm{out}}(t)$ beat frequencies [ref. Supplimentary], respectively, where $\chi_{\mathrm{in}}(t)$ and $\chi_{\mathrm{out}}(t)$ represent frequency excursions acquired by the SL resulting from environmental disturbance coupled to the sensing fibre, whereas, the PCF by virtue of its working ideally does not add any noise to it [ref. to Stanley paper]. The single passed SL through PCF $\rightarrow$ sensor fibres (out-of-loop) producing $\delta\nu_{\mathrm{out}}(t)$ experience the external perturbations once while propogates through the sensing fibre, whereas, the PCF $\rightleftharpoons$ Sensor double passed (in-loop) producing $\delta\nu_{\mathrm{in}}(t)$ experiances twice of that perturbation due to its to-and-fro propagation. Experiencing twice the sound wave by the SL is possible since light propagates at a six orders of magnitude faster speed than sound. However, in practice, the efficiency of the PCF differs from being an ideal phase-null stabilization [ref. to Stanley paper], which we consider in our analysis. The two beatnotes are then recorded using a frequency counter [50] or an oscilloscope.

The intrinsic phase $\Phi_L(t)$ of the SL passing through the sensing fibre is affected by any environmental perturbations and changed to $\Phi_P(t)$, which, therefore, can be sensed by accurate measurement of the change in SL's phase $\Phi(t) = \Phi_P(t) - \Phi_L(t)$. For a detailed understanding of the underlying mathematical modeling, readers are referred to Ref. [ref. supplementary]. We consider a sinusoidal acoustic source $S(f_S,t) = S_\circ \sin(2\pi f_S t + \phi_S)$ having amplidute $S_\circ$, frequency $f_S$ and at an arbitrary phase $\phi_S$ originating

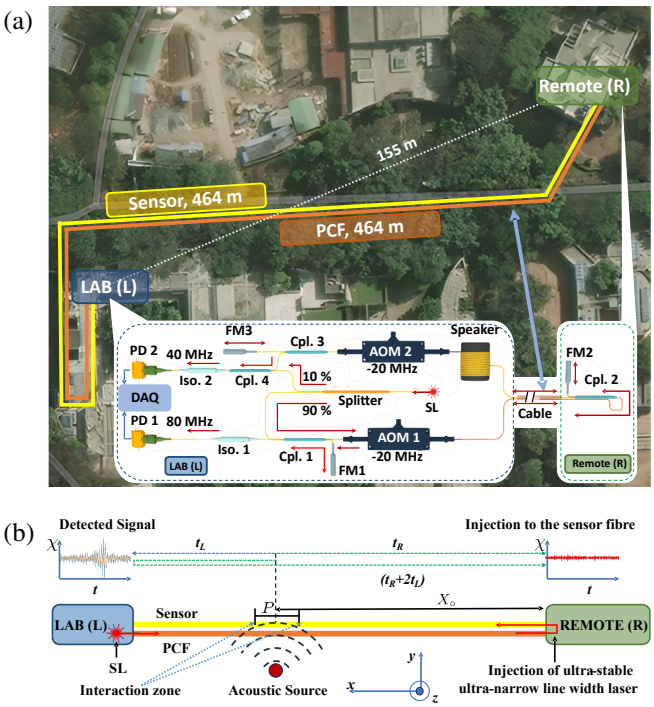



Fig. 1. (a) Optical arrangement together with out-of-the-lab sensor and PCF fiber links connecting two distant locations, lab (L) and remote (R). Here, optical frequency reference (OFR), Coupler (Cpl), acousto optic modulator (AOM), Faraday mirror (FM), photodiode (PD),and isolator (Iso). (b) SL-DAS schematic showing the time-delayed propagation of the in-loop and the out-of-loop signals upon experiencing any instantaneous acoustic signal.

at an unknown position thereafter propogating through the environmental medium and impinging on the sensing fibre at a position $P(X_\circ)$ that is $X_\circ$ away from R (Fig. 1b), which is resulting a frequency excursion $\chi(t) = (2\pi)^{-1}\, d\Phi(t)/dt$ to the SL's frequency. The SL propagates through the sensing fibre (SMF-28 Ultra) at a speed $c_f = c/n_{\mathrm{eff}}$, where $c$ is light speed in vaccum and $n_{\mathrm{eff}} = 1.462$ is the effective group index of refraction of the fiber core at $1550\,\mathrm{nm}$. The in-loop circulation of SL from $P \rightarrow R$ followed by $R \rightarrow P \rightarrow L$ occurs at a travel time $t_R$ and $t_R + 2t_L$, respectively. Therefore, the in-loop circulating SL reaching at L is twice affected by all the acoustic sources $S(f_{S,i},t)$ impinging on the fiber at different locations $\mathrm{P}_i$, and the resulting excursion is given as,

$$\chi_{\mathrm{in}}(t) = \sum_i 2\chi_{\circ,i} \sin\big(2\pi f_{S,i}(t - t_{L,i} - t_{R,i}) - \phi_{S,i}\big) \times \cos\big(2\pi f_{S,i} t_{L,i}\big), \quad (1)$$

whereas, the out-of-loop circulation from $P \rightarrow L$ taking time $t_L$ experiences $S(f_{S,i},t)$ once and the excursion is represented as,

$$\chi_{\mathrm{out}}(t) = -\sum_i \chi_{\circ,i} \sin\big(2\pi f_{S,i}(t - t_{L,i}) - \phi_{S,i}\big). \quad (2)$$

Two recorded beat notes $\delta\nu_{in}(t)$ and $\delta\nu_{out}(t)$ are further processed [ref. supplementary] to extract the mean phase difference $\langle\Delta\phi_i\rangle$ between the in-loop and out-of-loop waves for each acoustic frequencies $f_{S,i}$, which is used to determine the mean location $\mathrm{P}_i(\bar{X}_{m,i})$ where the $f_{S,i}$ impinges on the

fiber as,

$$\bar{X}_{m,i} = \frac{c}{n_{\text{eff}}} \cdot \frac{\langle \Delta\phi_i \rangle}{2\pi f_{S,i}}. \quad (3)$$

Accuracy in determining the location of $P_i$ that is $\Delta X_{m,i}$ depends on the measurement inaccuracy of $f_{S,i}$ and the standard deviation of $\langle \Delta\phi_i \rangle$. Unlike the $\phi-$OTDR based DAS, the SL-DAS detects beat frequencies proportional to the instantaneous optical frequency shift of the probe laser rather than measuring the optical phase modulo $2\pi$; which therefore can be counted continuously without the modulo ambiguity, so the time derivative of phase (i.e. frequency) yields an unambiguous measure of perturbation-induced optical path length change [51]. The use of an SL enables the SL-DAS to perform a high-accuracy measurement due to its low $\Phi_L$ and mHz-level resolution capability in detecting $f_{S,i}$, which we studied through numerical simulations and also validated experimentally. For performing the simulation, we begin with a time-dependent laser frequency array resembling a CW laser, characterized by particular linewidths $\Delta\nu_L(t)$ and drifts $d\nu_L/dt$ [ref. supplementary]. Allan deviations (ADEV) for this probe laser's fractional frequency $\Delta\nu_L/\nu_L(t)$ are shown in Fig. 2(a) for a variety of laser parameters. Practically for any laser $d\nu_L/dt \neq 0$, however, the drift can be very low and linear by referencing the probe laser to an ultrastable Fabry-Perot cavity [52] and using low noise servo control [53]. On the other hand, the coherence property of a laser improves with its monochromaticity, i.e., ultra-narrow $\Delta\nu_L$, which is achieved by using a high finesse Fabry-Perot cavity. The probe laser used in our experiment corresponds to $\Delta\nu_L \approx 1\,\text{Hz}$ and $d\nu_L/dt = 33\,\text{mHz/s}$, therefore has excellent coherence property and ultrastable during the sensing timescale (naming this a super-coherent laser). We then numerically mimic the SL-DAS, considering $f_S = 36\,\text{Hz}$ located at $X_\circ = 450\,\text{m}$ producing $\chi_{out} = 100\,\text{Hz}$, as an example, and the simulated accuracies $\Delta X_m$ in determining $X_\circ$ i.e. $\bar{X}_m$ are depicted in Fig. 2(b). This show betterment in $\Delta X_m$ can be achieved with narrower $\Delta\nu_L$; however, it does not depend on $d\nu_L/dt$ as long as the propagation time, $\tau$, through the sensing fibre satisfies $\tau \cdot d\nu_L/dt << \Delta\nu_L$. Therefore, for the SL-DAS, a low and linear $d\nu_L/dt$ SL is suitable for scaling up of the sensing length since the coherence length increases with a narrower $\Delta\nu_L$ and coherence among the probe laser photons remains ensured [ref. supplementary].

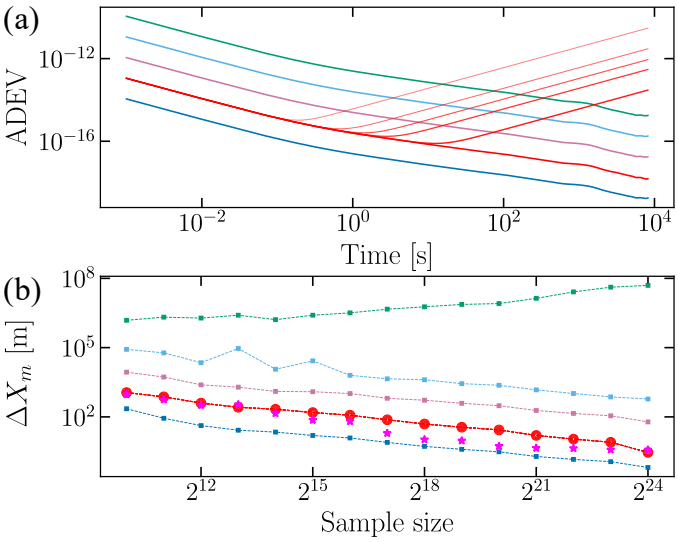



Fig. 2. (a) Alan deviation (ADEV) for a set of undrifted CW lasers having linewidth $\Delta\nu_L = 1$ kHz (green), 0.1 kHz (cyan), 10 Hz (purple), 1 Hz (red), and 0.1 Hz (blue). ADEV of the laser having $\Delta\nu_L = 1$ Hz is extended considering linear drifts $d\nu_L/dt = 1$ Hz/s, 100 mHz/s, 33 mHz/s (closely matching the SL used in the setup), 10 mHz/s, and 1 mHz/s (light to dark red). (b) The estimated accuracy $\Delta X_m$ in determining the position of an acoustic source impinging on the sensor fiber for all the different lasers (color scheme follows). $\Delta X_m$ for the laser with $\Delta\nu_L =$ Hz and $d\nu_L/dt = 1 - 1000$ mHz are distinguished by smaller to larger circle diameters (red), therefore, independent of $d\nu_L/dt$. The experimental result for the SL used in SL-DAS is also indicated (magenta).

## III. Results And Discussion

The beat frequencies $\delta\nu_{in}(t)$ and $\delta\nu_{out}(t)$ are recorded at a 1 kHz sampling rate and the ac components in these, $\chi_{\text{in}}(t)$ and $\chi_{\text{out}}(t)$, as shown in Fig. 3(a), are relevant to extract the desired acoustic or seismic information [ref. supplementary]. Fast Fourier Transform (FFT) of $\chi_{\text{in}}(t)$ and $\chi_{\text{out}}(t)$ generates various $f_{S,i}$ components in these (Eqs. 1 and 2). To benchmark the SL-DAS, we externally inject eleven known distinct frequencies with $f_{S,i} = 31 : 1 : 41\,\text{Hz}$ at a given location $P(X_\circ = 450\,\text{m})$ to the sensor fibre simultaneously by using a speaker, where the acoustic strength $S_{\circ,i}$ of these can be group-controlled such that the resulting $\chi_{out,i} = \kappa_i S_{\circ,i}$ can be varied from 0.1 Hz to 200 kHz for $\kappa_i$ being the coupling factor effective to the probe laser for each $f_{S,i}$. The effective coupling factors $\kappa_i$, which depend on source strength and their relative separation from the sensor fiber, are enhanced for these markers by wrapping 15 turns of a 2 m fibre on the speaker. This way, their resulting perturbations on the SL are boosted and distinguish them from any environmental residuals that match the marker frequencies. The detected markers, along with other available frequencies, are shown in Fig. 3(b), where different excursions at all the markers are an artifact of the speaker producing a frequency-dependent $S_{\circ,i}$. From this, the measured value of $\chi_{\text{in}}/\chi_{\text{out}}$ is $1.999946(27)$, therefore, confirms an efficient operation of the PCF, which ideally has to be 2. To understand the underlying phenomena, we modeled the envelope of $\chi_{in}$, as shown in Fig. 3(a), by parameterizing the phase matching among all $\phi_{S,i}$, once considering only the markers, then considering all acoustic components generated from FFT (Fig. 3(b)). The latter one undoubtedly explains the envelope better, giving rise to confidence in the SL-DAS detection and our understanding. Figure 3(c) shows the sensitivity spectrum of the SL-DAS, i.e., the system noise essentially contributed by optics and electronics captured using a 1 m sensor fibre for different integration times. This is about a factor of three better than the best reported optical fiber-based DAS result [54]. Figure 3(d) shows detection as low as $f_S = 0.1\,\text{Hz}$ along with 1 Hz and 1 kHz acoustic signals, as a few examples. This measured bandwidth is limited by capability of the actuators available with us, even though the SL-DAS is capable of sensing from 0.1 mHz to $> 2\,\text{kHz}$. In comparison, commercial mechanical sensors

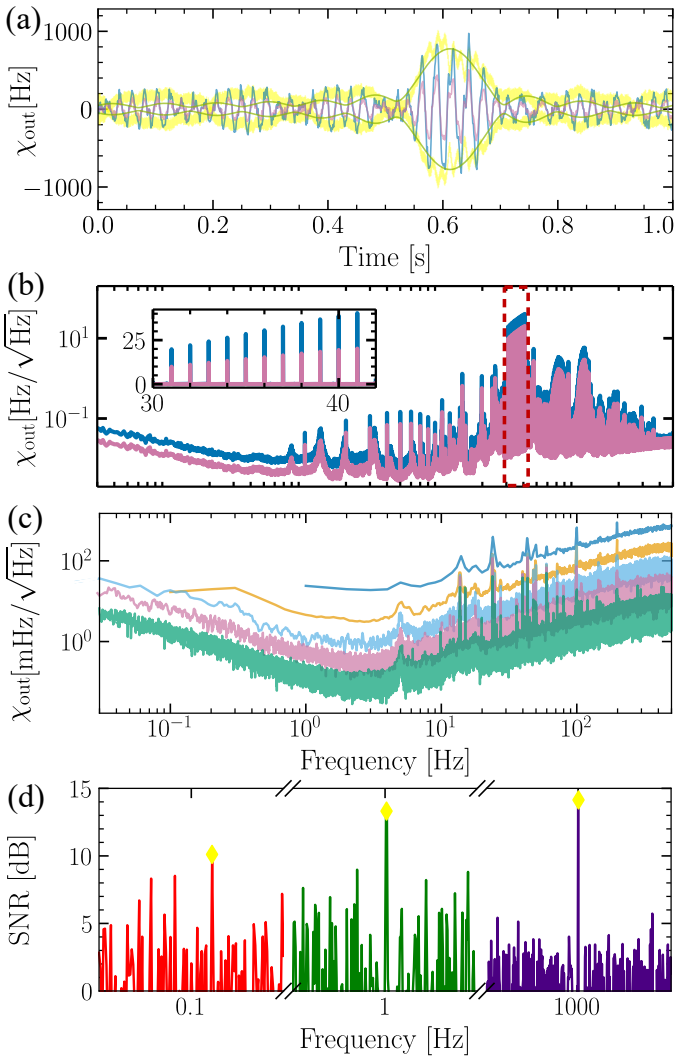



Fig. 3. Out-of-loop $\chi_{out}$ (magenta), and in-loop $\chi_{in}$ (blue) typical recorded data: (a) in the time domain, where the envelop is modeled by phase matching of the traveling acoustic waves considering only the markers (gray), and all frequencies (yellow); (b) in the frequency domain, where markers are segregated in a box (red-dashed) and their zoomed view (inset). (c) System noise recorded in the out-of-loop signal using a 1 m sensing fiber for sampling times of 1 s (blue), 10 s (yellow), $10^2$ s (cyan), $10^3$ s (magenta), and $10^4$ s (green). (d) Detection of 0.1 Hz (red), 1 Hz (green), and 1 kHz (violet) acoustic frequencies, which are produced using external shakers, where the respective peaks are indicated (yellow).

operate only above 10 mHz [55] that too without having distributed sensing capability. To demonstrate the distributed nature of the SL-DAS, we extended the sensing length using fiber spools and placed two speakers at different locations $P_1(X_\circ^1 = 6\,\text{m})$ and $P_2(X_\circ^2 = 9233\,\text{m})$, respectively, which are then operated simultaneously at different frequencies $f_S^1$ and $f_S^2$, respectively. We tuned both $f_S^1$ and $f_S^2$ in steps within a band $53 - 95$ Hz such that $f_S^1$ is changed in increasing order, whereas $f_S^2$ is in decreasing order, and in the midway $f_S^1 = f_S^2 = 83$ Hz they match. Figure 4 shows the ability of the SL-DAS detecting $f_S^1$ and $f_S^2$ simultaneously, therefore clearly demonstrates its distributed sensing capability. We distinctly detect their individual mean positions $\bar{X}_m^1 = 9.4 \pm 10.4\,\text{m}$

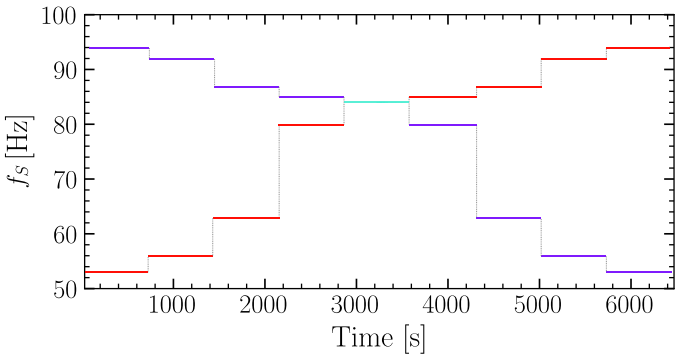



Fig. 4. SL-DAS's ability for distributed sensing: determining positions $\bar{X}_m^1$, and $\bar{X}_m^2$ of the two external acoustic sources placed at $P_1(X_\circ^1 = 6\,\text{m})$ (red), and $P_2(X_\circ^2 = 9233\,\text{m})$ (purple), respectively, and operated simultaneously at different frequencies except in their midway of tuning when both produce 83 Hz (cyan).

and $\bar{X}_m^2 = 9231.6 \pm 10.7\,\text{m}$, except when both the speakers operate at the same frequency of 83 Hz. The large uncertainties associated with $\bar{X}_m^1$ and $\bar{X}_m^2$ appear due to cross-talk that becomes prominent while using the fiber spool and strong $S_\circ$, which was deliberate so that we operate SL-DAS at a smaller sample size for each $f_S^1$ and $f_S^2$ since the underlying motivation is to demonstrate its distributed sensing nature only. It is observed that the artifact of cross-talk disappears in a geographically distributed fiber. During the simultaneous occurrence of identical frequencies in different locations, the probe SL acquires a cumulative $\langle \Delta\phi \rangle$ resulting from those, which are indistinguishable by the FFT; therefore, the SL-DAS sense their average location (Fig. 4). In the current setup, with the available laser power, we demonstrated a sensing length of up to 80 km (using spools), which is not limited to, and is approximately twice that of any other optical fiber-based DAS [56] without using special techniques, e.g., scattering centers, or bi-directional repeaters.

Upon benchmarking the SL-DAS, we operated it in an out-of-lab environment using a 464 m long field-deployed sensor fibre connecting two different buildings, as shown in Fig. 1(a). The recorded spectrogram is depicted in Fig. 5(a), showing detection of acoustic frequencies from unknown sources along with the markers placed at $X_\circ = 459\,\text{m}$ and their harmonics. The observed stray frequencies, originating from unidentified sources, have varying strengths and appear either as a pulse or for a certain duration; some of them are sequential, while others are present throughout the data-taking duration. Identifying the respective sources and analyzing their properties is not the focus of this article; however, a 50 Hz signal originating from the AC power lines (Indian standard) and present at all times is easily identifiable. Figure 5(b) shows measured locations, $\bar{X}_m \pm \Delta X_m$ for some of these detected frequencies along with all the markers for a fixed sample size of $2^{18}$ and each of which is integrated over 1 ms. All the markers originating at $X_\circ = 459\,\text{m}$ are measured to be at average location of $\bar{X}_m = 459.4\,\text{m}$. However, each of their individual $X_{m,i}$ statistically differs from $\bar{X}_m$ with a varying $\Delta X_{m,i}$ since the speaker produce them at different strengths $S_{\circ,i}$ (inset

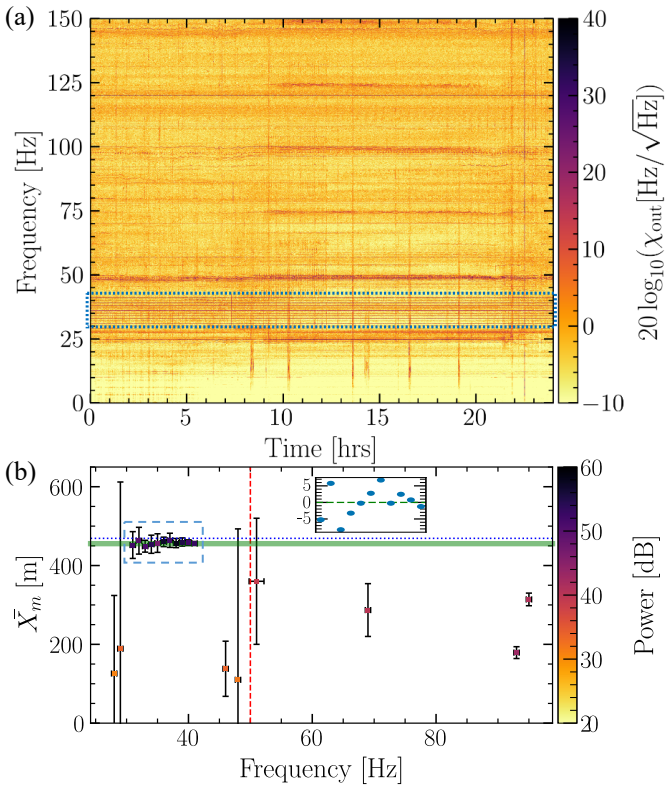



Fig. 5. (a) An example spectrogram of the SL-DAS detected acoustic sources where the markers are segregated in a box (blue dashed). (b) Measured locations $\bar{X}_m$ for some of the unidentified sources, along with markers (blue dashed box). Stronger source locations are better position resolved; otherwise, they impinge on longer lengths of the sensing fiber, whereas frequency error bars indicate the source bandwidth. $\bar{X}_m$ of the markers (inset), even though they are generated at a fixed $X_\circ$, differs since they are produced at varying strengths by the speaker.

Fig. 3(b)). For the others, $\Delta X_{m,i}$ varies even though all of them are detected for a fixed sample size, due to their different strengths, which is explained in the later section. Additionally, if the waves interact over a longer length with the sensing fiber, typically either for a far-away source or an elongated source, they also result in a larger $\Delta X_{m,i}$.

The SL-DAS detected resolution in $f_S$ and signal-to-noise ratio (SNR) improves with increasing sample size, as shown in Fig. 6, which is obtained using a marker at $f_S = 36\,\text{Hz}$ and setting its strength such that $\chi_{out} = 15.3\,\text{Hz}$. Experimental results are also explained through theoretical modeling, considering a 36 Hz single-frequency sinusoidal acoustic source. For this we consider $\chi_\circ = 15.3\,\text{Hz}$, $\phi_S = 0$, $t_L$ and $t_R$ corresponding to $\text{P}(X_\circ = 459\,\text{m})$ to construct the corresponding $\chi_{out}(t)$ array for $t$ being 0 to different sampling time (i.e., sample size $\times$ 1 ms), following Eq. 2. Instead of $f_S$ being an ideal monochromatic, rather, practically having a 1 mHz linewidth due to random noise, matches the experimental data (Fig. 6). The minimum required sample size to achieve the maximum achievable resolution is independent of the signal strength and depends only on the sampling time, as determined by our model and verified through experimental observations as well.

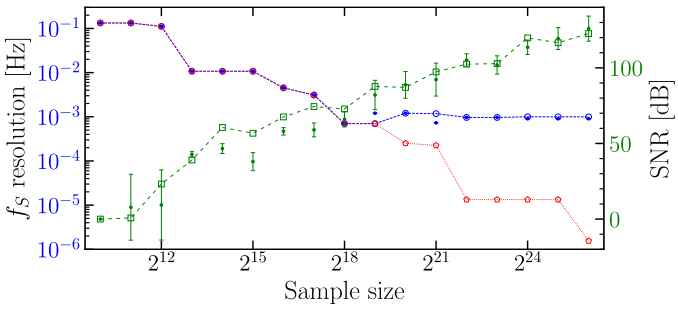



Fig. 6. Experimentally measured (indicated by filled markers) frequencies (red) are better resolved, and SNR (green) improves with an increasing sample size. The same features are also verified through simulations (indicated by unfilled markers). In practice, $f_S$ is not monochromatic; instead, it matches the measurement for 1 mHz intrinsic linewidth (blue). The experiment and the simulations are carried out for identical conditions, i.e., a marker of 36 Hz producing a fixed excursion of 15.3 Hz, as an example.

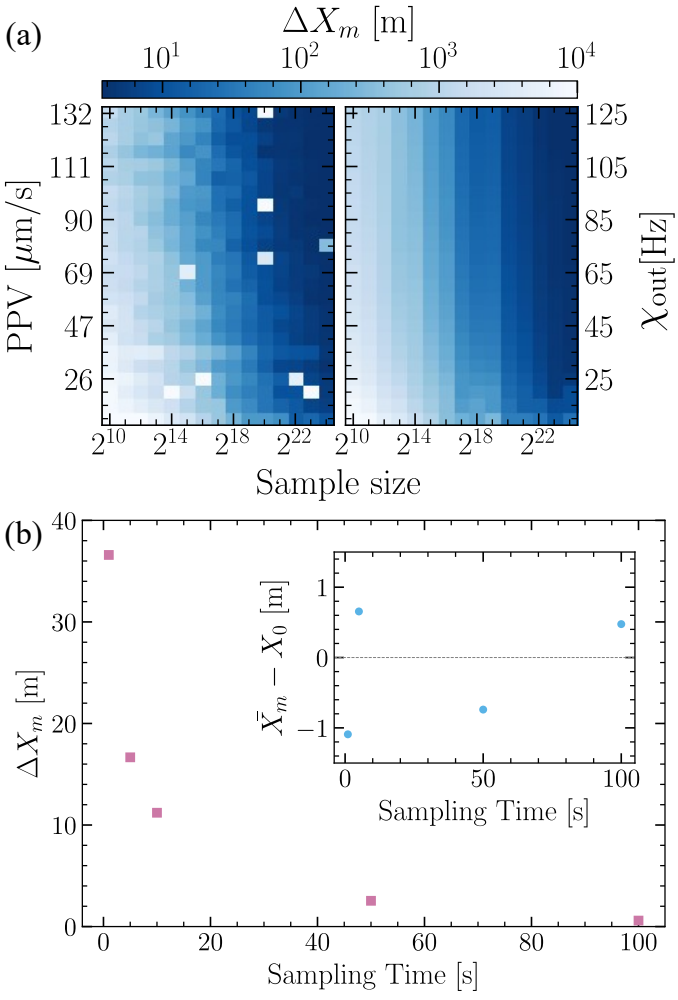



Fig. 7. (a) Experimentally measured (left) and simulated (right) $\Delta X_m$ in determining $\bar{X}_m$, which essentially reaches to the system limited resolution. A smaller sample size can localize to its best limit for strong sources and vice versa. (b) The characteristic timescale of localization for a loud source producing a 1 kHz excursion at a 36 Hz marker, for example, is examined while placing the speaker at 918 m, 9.2 km, 21.3 km, and 22.7 km for consistency checks. Results for all four lengths overlap in the scale of this plot. Deviations of their average detected locations from the actual speaker position are shown in the inset.

It is experimentally observed that the position $P(X_\circ)$ can be accurately localized even using a small sample size when $\chi_{out}$ is strong; however, a large sample size is required for a weak signal. In the experiment, a marker at $f_S = 36\,\mathrm{Hz}$ is placed at $X_\circ = 459\,\mathrm{m}$ in the deployed fibre; $\chi_{\mathrm{out}}(t)$ and $\chi_{\mathrm{in}}(t)$ are recorded over several hours for each different excursions that is varied by the speaker's strength. From each time series excursion data for a fixed strength, random sets are selected for $2^n$ sample size for $n = 10 : 1 : 24$ to estimate the corresponding $\bar{X}_m$ and associated standard deviation $\Delta X_m$ for each $n$. This is performed for all recorded $\chi_{\mathrm{out}}(t)$ and $\chi_{\mathrm{in}}(t)$, and the obtained results are depicted in Fig. 7(a-left). To perform the analogue simulation, we begin by creating time arrays of in-loop and out-of-loop circulating laser frequencies $\nu_L^{\mathrm{in}}(t)$, and $\nu_L^{\mathrm{out}}(t)$, respectively, perturbed by the external sources [ref. supplimentary]. For this, $\chi_{\mathrm{in}}(t)$ and $\chi_{\mathrm{out}}(t)$ are experimentally measured values and the additional broadening-cum-drift of the probe laser $\Delta\nu_{PCF}$ due to its travel through PCF is constructed considering the laser parameters $\Delta\nu_L = 1\,\mathrm{Hz}$ and $d\nu_L/dt = 33\,\mathrm{mHz/s}$ matching to the SL used in this experiment. Therefore, the measured excursions also contain an additional $\Delta\nu_{PCF}$, which are sampled to a $2^n$-scale in a similar way the experimental data is processed, and the calculated $\Delta X_m$ values are depicted in Fig. 7(a-right). Both the experiment and simulation match each other and show a minimum resolvable $\Delta X_m = \pm 0.85\,\mathrm{m}$, which is a factor of three improvement compared to any other reported optical fiber based DAS implemented on deployed fiber [22]. To understand the effectiveness of an SL, we repeated the simulations for a variety of laser parameters, and the experimental result matches very well with the simulation at the SL parameter, as shown in Fig. 2. This shows that $\Delta\nu_{PCF}$ has a negligible effect on the SL-DAS resolution while using an SL, but deviates largely for a non-supercoherent laser. This not only builds confidence in the numerical simulations but, more importantly, shows the importance of the SL. Further, for the consistency check, $\chi_{out}$ is kept fixed to $1\,\mathrm{kHz} \sim 1\,\mathrm{mm/s}$ of PPV; and $\Delta X_m$ is extracted for different sampling times ($2^n \times 1\,\mathrm{ms}$), which is then repeated by changing the marker position $X_\circ$ to 918 m, 9.2 km, 21.3 km, and 22.7 km by adding fiber spools. The measured $\Delta X_m$ for all different $X_\circ$ are shown in Fig. 7(b), and their indistinguishability (in this scale) confirms the scalability of the SL-DAS. Above a threshold sampling time for this given $\chi_{out}$, the minimum $(\bar{X}_m - X_\circ)$ is obtained to be $0.5\,\mathrm{m}$, which is well within the SL-DAS resolution of $\Delta X_m = \pm 0.85\,\mathrm{m}$.

We extract the peak particle velocity $\nu$, displacement $\mathcal{D}$, and strain $\epsilon$ those characterize seismic and acoustic waves using SL-DAS measured $\chi_{out}(t)$, and $\chi_{in}(t)$ [ref. supplimentary]. External shakers are used to produce excursions at their lowest possible level at various frequencies, ranging from 0.1 to 1000 Hz, which is limited by the shaker's capability, and the results are shown in Fig. 8(a-b). The lowest sensitivities of the SL-DAS are determined to be $\nu = 2.4\ \mathrm{nm/s}$, $\mathcal{D} = 380\ \mathrm{fm}$, and $\epsilon = 190 \times 10^{-15}$ measured at $f_S = 1\ \mathrm{kHz}$, which is about a factor of three improvement compared to the earlier reported $560 \times 10^{-15}$ in a deployed optical fiber based DAS [54]. The SL-DAS is capable of detecting $f_S$ for $\chi_{out} \geq 4\ \mathrm{mHz}$ using

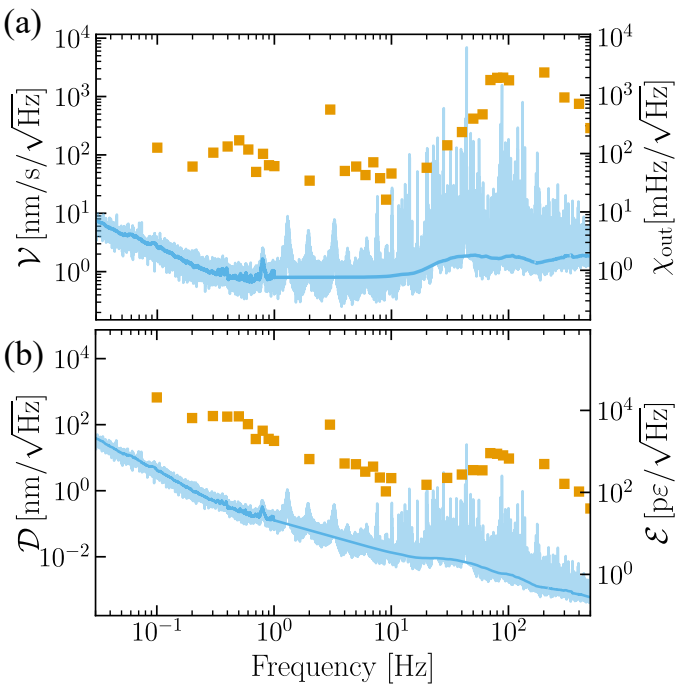



Fig. 8. System limited detection sensitivity of SL-DAS (blue), along with residual acoustics (light blue), measured using the field-deployed fiber. External shakers are used at their lowest excursion levels over a bandwidth of 0.1 Hz to 1 kHz, and properties (a) peak particle velocity $\nu$, (b) displacement $\mathcal{D}$, strain $\epsilon$, characterizing the waves are shown.

FFT, whereas, their position $P(\bar{X}_m)$ determination requires a minimum $\chi_{out}$ of 9 Hz whose accuracy gets improved for stronger $\chi_{out}$. The sensing bandwidth of the SL-DAS is $0.1\ \mathrm{mHz} - 1\ \mathrm{kHz}$; however, availability-limited acoustic signals are detected $\geq 0.1$ Hz, whereas slow temperature variations are detected at the lower frequencies.

## IV. Conclusion

The sensitivity of the SL-DAS is enhanced by utilizing a supercoherent laser, which delivers unprecedented sensitivity, source frequency, and position resolutions over the range of $0.1\ \mathrm{mHz}$ to $1\ \mathrm{kHz}$, thereby setting a new benchmark for distributed sensing. The system employs a dual self-heterodyne scheme with a nearly monochromatic, ultra-low linearly drifting continuous-wave SL, rendering the sensor inherently drift-free in its operation. The demonstrated operational length is 80 km, which can be scaled up by using higher SL power and single-photon avalanche detectors, without requiring any bi-directional repeaters. This holds great promise for perimeter and underwater surveillance, in addition to the typical DAS applications. Another potential use of the SL-DAS is in gravitational-wave detectors, where it provides accurate detection of seismic and acoustic disturbances, especially at low frequencies, enabling the real-time acquisition of effective noise mitigation coupled into the interferometer arms.

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# Supplementary

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## A. Theory of Supercoherent Laser-based Distributed Acoustic Sensing (SL-DAS)

In this section, we are describing the detailed mathematical formulation of the super-coherent photon based ultra-sensitive Distributed Acoustic Sensing experiment. The electromagnetic formulation of the super-coherent laser (SL) at a frequency $\nu_L$ is,

$$E(t) = E_\circ e^{i(2\pi\nu_L + \Phi_\circ)}, \tag{1}$$

where $E_\circ$ is the amplitude $\Phi_\circ$ is the intrinsic phase noise of SL. In the experiment, the SL makes single and double pass through two AOMs 1 and 2 for the out-of-loop and in-loop circulations, respectively, as shown in Fig. 1(b) in the manuscript. Each AOM during a single pass of the laser, downshifts $\nu_L$ by 20 MHz. Therefore, after traveling through an unperturbed sensing optical fibre of length $L_\circ$, the modified laser frequency for in-loop circulation is,

$$\nu_L^{\text{in}} = \nu_L - 80\,\text{MHz}, \tag{2}$$

and for the out-of-loop circulation it becomes,

$$\nu_L^{\text{out}} = \nu_L - 40\,\text{MHz}. \tag{3}$$

Readers are referred to read the experiment section in the manuscript for a better understanding of the in-loop and out-of-loop circulations.

However, in practice, coupling of the environmental noises to the sensing fibre induces local microscopic length changes to $L_\circ$. Considering a sinusoidal disturbance of strength $S_\circ$ with frequency $f_S$ and at an arbitrary phase $\phi_S$, originating at an unknown location then upon propagating it is impinging to the fibre with a coupling constant $\kappa$, resulting to its length change,

$$\begin{aligned}\Delta L_S(t) &= \Delta L_\circ \sin(2\pi f_S + \phi_S) \\ &= \kappa S_\circ \sin(2\pi f_S + \phi_S),\end{aligned} \tag{4}$$

where $\Delta L_\circ$ is the maximum length change of the fibre due to the peak local displacement caused by the disturbance. The total length change in $L_\circ$ is the accumulation of effects from all acoustic sources $f_{S,i}$, where $i$ depicts the $i^{\text{th}}$ source, and is represented as,

$$\begin{aligned}L(t) &= L_\circ + \sum_i \Delta L_{S,i}(t) \\ &= L_\circ + \sum_i \Delta L_{\circ,i} \sin(2\pi f_{S,i} t + \phi_{S,i}).\end{aligned} \tag{5}$$

Therefore, accumulation of phase upon propagation of SL through the perturbed fibre is,

$$\Phi(t) = \frac{2\pi n_{\text{eff}} L(t)}{\lambda_L}, \tag{6}$$

where $\lambda_L$ is the wavelength of the SL in vacuum, and $n_{\text{eff}}$ is the group refractive index of the fibre core. This resultant phase change $\Phi(t)$ in SL, carries relevant information about the perturbation and is therefore used as the basis for the sensing. We use self-heterodyne, where an unperturbed local copy of SL, $E_L^R(t) = E_\circ^R e^{\Phi_L(t)}$ is being interfered with its counterpart that has travelled through the sensing fibre $E_L^P(t) = E_\circ^P e^{\Phi_P(t)}$ impinging on a fast photodiode and the resultant photocurrent is,

$$I(t) \propto 2E_\circ^R E_\circ^P \cos\big(\Phi(t)\big), \tag{7}$$

where $E_\circ^R$ and $E_\circ^P$ represent amplitudes of the reference and probe, respectively, and

$$\begin{aligned}\Phi(t) &= \Phi_P(t) - \Phi_L(t) \\ &= \frac{2\pi n_{\text{eff}}}{\lambda_L}\left[L_\circ + \sum_i \Delta L_{\circ,i} \sin(2\pi f_{S,i} t + \phi_{S,i})\right]\end{aligned} \tag{8}$$

Therefore, the resulting frequency excursion is,

$$\chi(t) = \frac{1}{2\pi}\frac{d\Phi(t)}{dt} \tag{9}$$

and using $\Phi(t)$ from Eq. 8, it further reduces to,

$$\begin{aligned}\chi(t) &= \sum_i \left(\frac{2\pi n_{\text{eff}} f_{S,i} \Delta L_{\circ,i}}{\lambda_L}\right) \cos\big(2\pi f_{S,i} t + \phi_{S,i}\big) \\ &= \sum_i \chi_{\circ,i} \sin\Big(\tfrac{\pi}{2} - (2\pi f_{S,i} t + \phi_{S,i})\Big) \\ &= -\sum_i \chi_{\circ,i} \sin\big(2\pi f_{S,i} t - \phi_{S,i}\big)\end{aligned} \tag{10}$$

where,

$$\chi_{\circ,i} = \frac{2\pi n_{\text{eff}} f_{S,i} \Delta L_{\circ,i}}{\lambda_L}, \tag{11}$$

is the excursion from $f_{S,i}$ of the external noise. In a practical scenario, fibres always experience the external perturbations; therefore, upon circulation of SL through the in-loop and out-of-loop as given in Eq. 2 and 3, respectively, are modified by the excursion (Eq. 10) resulting from the perturbed fibres. In case of in-loop circulation, SL travels $P_i \to L \to P_i \to R$ consisting of two parts $P_i \to L$ followed by $L \to P_i \to R$ (Fig. 1(b) in the manuscript), where $P_i$ denotes the position where the acoustic source with $f_{S,i}$ impinges on the sensing fibre. Therefore, in this in-loop detection channel, SL experiences twice the excursions from $f_{S,i}$ and the laser frequency

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reaching PD 1 is,

$$\nu_L^{\text{in}}(t) = \nu_L^{\text{in}} + \sum_i \chi_{i,P_i\to L}(t) + \sum_i \chi_{i,L\to P_i\to R}(t) + \Delta\nu_{\text{PCF}}(2\cdot\tau_{\text{PCF}}), \tag{12}$$

whereas for the out-of-loop $P_i \to L$ circulation, SL experience once the excursion from $f_{S,i}$ and the laser frequency reaching PD 2 is,

$$\nu_L^{\text{out}}(t) = \nu_L^{\text{out}} + \sum_i \chi_{i,P_i\to L}(t) + \Delta\nu_{\text{PCF}}(\tau_{\text{PCF}}). \tag{13}$$

Here, the additional broadening-cum-drift of the probe laser upon travelling through the PCF at a time $\tau_{\text{PCF}}$ is given as,

$$\Delta\nu_{\text{PCF}}(\tau_{\text{PCF}}) = \Delta\nu_L \cdot a(\tau_{\text{PCF}}) + \frac{d\nu_L}{dt}\cdot\tau_{\text{PCF}}, \tag{14}$$

where $a$ is the broadening factor to $\nu_L$ resultant from a non-ideal PCF that also depends on the length of the PCF. Here, $\Delta\nu_{\text{PCF}}(2\cdot\tau_{\text{PCF}}) = 2\cdot\Delta\nu_{\text{PCF}}(\tau_{\text{PCF}})$, since PCF is very stable over timescale $\gg \tau_{\text{PCF}}$, therefore the resultant to-and-fro broadening of the probe laser is twice that of traveling along one direction. Additionally, when using an SL as the probe laser, the resulting drift $\frac{d\nu_L}{dt}\cdot\tau_{\text{PCF}}$ is low. In a self-heterodyne technique, the in-loop beat frequency detected at PD 1 is,

$$\begin{aligned}\delta\nu_{\text{in}}(t) &= \nu_L - \nu_L^{\text{in}}(t)\\ &= 80\text{ MHz} - \Big[\sum_i \chi_{i,P_i\to L}(t) + \sum_i \chi_{i,L\to P_i\to R}(t)\Big]\\ &\quad - 2\cdot\Delta\nu_{\text{PCF}}\cdot a(\tau_{\text{PCF}}),\end{aligned} \tag{15}$$

and at out-of-loop at PD 2 is,

$$\begin{aligned}\delta\nu_{\text{out}}(t) &= \nu_L - \nu_L^{\text{out}}(t)\\ &= 40\text{ MHz} - \sum_i \chi_{i,P_i\to L}(t) - \Delta\nu_{\text{PCF}}\cdot a(\tau_{\text{PCF}}),\end{aligned} \tag{16}$$

upon replacing $\nu_L^{\text{in}}(t)$ and $\nu_L^{\text{out}}(t)$ as given in the Eq. 12 and 13, respectively. In case of using a linearly drifting probe laser, as the SL used in the SL-DAS, $\frac{d\nu_L}{dt}\cdot\tau_{\text{PCF}}$ appears as a dc offset in both the reference and in/ out-of-loop circulating lasers in the self-heterodyne detection; therefore, it cancels. However, this can not be neglected for any standard laser systems. Whereas the use of a PCF link ensures $\Delta\nu_{\text{PCF}}\cdot a(\tau_{\text{PCF}} < \chi_{\text{out}}$ for an SL, therefore, can be neglected for simplicity.

In a self-heterodyne, the SL plays an important role due to its large coherence length $L_C$ resulting from a sub-Hz linewidth whereas the low phase noise laser attributes common mode rejection of the laser's intrinsic noise between the short (reference) and delayed (sensing) paths. The RF beat signals, $f_{\text{RF}}$ are filtered using proper band-pass filters and sent to a precision frequency counter, which produces two time series signals, which are recorded for further processing and data analysis. For the in-loop channel upon removing the 80 MHz offset we get,

$$\chi_{\text{in}}(t) = \sum_i \chi_{i,\,P_i\to L}(t) + \sum_i \chi_{i,\,L\to P_i\to R}(t), \tag{17}$$

and then using Eq. 10 it further reduces to,

$$\begin{aligned}\chi_{\text{in}}(t) &= -\sum_i \chi_{\circ,i}\sin\big(\phi_{S,i} - 2\pi f_{S,i}(t - t_{L,i})\big)\\ &\quad -\sum_i \chi_{\circ,i}\sin\big(\phi_{S,i} - 2\pi f_{S,i}(t - t_{L,i} - t_{R,i})\big)\\ &= -\sum_i 2\chi_{\circ,i}\sin\big(2\pi f_{S,i}(t - t_{L,i} - t_{R,i}) - \phi_{S,i}\big)\\ &\quad\times\cos\big(2\pi f_{S,i}t_{R,i}\big),\end{aligned} \tag{18}$$

Similarly for the out-of-loop channel upon removing 40 MHz offset we get,

$$\begin{aligned}\chi_{\text{out}}(t) &= \sum_i \chi_{i,\,P_i\to L}(t)\\ &= -\sum_i \chi_{\circ,i}\sin\big(2\pi f_{S,i}(t - t_{L,i}) - \phi_{S,i}\big).\end{aligned} \tag{19}$$

Here, $t_{L,i}$ and $t_{R,i}$ represent the travel time of photons from $P_i$ to the lab and to the remote, respectively. Figure 3(a) in the manuscript shows recorded $\chi_{\text{in}}(t)$ and $\chi_{\text{out}}(t)$, as an example. Both the RF beats $\delta\nu_{in}(t)$ and $\delta\nu_{out}(t)$ are measured by a reciprocal, interpolating counter (k+k), disciplined by an external 10 MHz reference from a rubidium (Rb) frequency standard, therefore the Alan deviation of the k+k counter nearly follows the Rb standard ($\simeq 10^{-11}$ at 1 s). The counter reports the finite-gate time average of $\delta\nu_{in}(t)$ and $\delta\nu_{out}(t)$ over preset integration time $T$ as,

$$\begin{aligned}F_{\text{meas}}(t_k) &= \frac{1}{T}\int_{t_k}^{t_k+T} f_{\text{RF}}(t)dt\\ &= f_{\text{nom}} + \frac{1}{T}\int_{t_k}^{t_k+T}\delta f_{\text{RF}}(t)dt,\end{aligned} \tag{20}$$

where $t_k$ is the start time and $f_{\text{nom}} \in \{40, 80\}$ MHz. Throughout the measurements $T = 1$ ms, which is the minimum allowance by the counter. In the frequency domain, this is equivalent to rectangular averaging with a measurement filter

$$H(f) = \text{sinc}(\pi f T)e^{-i\pi f T}, \tag{21}$$

therefore,

$$|H(f)|^2 = \text{sinc}^2(\pi f T), \tag{22}$$

which roll off above $\sim 1/T$. Therefore, the first zero at $1/T = 1$ kHz for $T = 1$ ms limits the highest detectable acoustic frequency in the present setup, which can be widened with a better counter. For 1 ms integration time the fractional frequency uncertainty is $\sim 10^{-8}$, which is negligible compared to SL-DAS resolution of detecting $\delta\nu_{in}$ and $\delta\nu_{out}$.

## B. Data Processing

Fast Fourier Transform (FFT) [1] of the acquired $\chi_{\text{in}}(t)$ and $\chi_{\text{out}}(t)$ as represented by Eq. 18 and 19 generates the corresponding $f_{S,i}$ following,

$$X(f_S) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi f_S t}dt, \tag{23}$$

where $x(t)$ represents, the excursion for each $f_{S,i}$. The Discrete Fourier Transform (DFT) is commonly computed using

the FFT algorithm, which efficiently computes the transformation with a complexity of $O(N \log N)$. Inverse Fourier Transform (IFT) given as,

$$x'(t) = \int_{-\infty}^{\infty} X(f_S) e^{j2\pi f_S t} df, \tag{24}$$

is used to reconstruct a time-domain signal from its frequency-domain representation as given in Eq. 23, where $x'(t)$ represents $\chi_i(t) = \chi_{0,i} \sin(2\pi f_{S,i} - \phi_{S,i})$ as given in Eq. 10. The Hilbert Transform is used to obtain the analytic signal that carries the information about its instantaneous amplitude and phase, which is given as,

$$\mathcal{H}\{x'(t)\} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x'(\tau)}{t - \tau} d\tau. \tag{25}$$

Here, $x'(t) \equiv \chi_i(t)$ comes from the measurement and $\tau$ represents the sampling time. Therefore the complete wave information of the acoustic signal $S(f_{S,i}, t) = \kappa \chi_i(t)$, for $\kappa$ being the coupling constant, is given as,

$$\mathcal{Z}(t) = \chi_i(t) + j\mathcal{H}\{\chi_i\}, \tag{26}$$

which is useful in demodulation and phase analysis [2]. Upon extraction of the phase of the acoustic signal, i.e., $\tan^{-1}\left(\frac{\mathcal{H}}{\chi_i}\right)$ its unwrapping is used to reconstruct the true phase when it is wrapped within a principal interval, such as $-\pi$ to $\pi$. By resolving phase ambiguities, unwrapping allows for extracting meaningful physical information from wrapped phase data which is particularly important in interferometry and spectral analysis [3]. Standard phase-sensitive distributed acoustic sensors measure the Rayleigh backscatter phase modulo $2\pi$. Therefore, when the true optical phase change exceeds $\pm\phi$, the measured phase wraps, leading to ambiguity that must be unwrapped or corrected; this is a well-recognized limitation in other DAS and can distort or clip strain/acoustic signals, especially under large perturbations or strong environmental noise, because phase unwrapping fails when phase jumps occur between samples [4]. In contrast, the SL-DAS detects the beat frequencies proportional to the instantaneous optical frequency shift rather than measuring the optical phase modulo $2\pi$; which can be counted or tracked continuously without any modulo ambiguity, so the time derivative of phase (i.e., frequency) yields an unambiguous measure of perturbation-induced optical path length change. As a result, SL-DAS inherently avoids the $2\pi$ phase wrapping problem because they measure frequency, i.e., the phase change rate, directly rather than wrapped phase, enabling large dynamic range detection provided the reference and counters are stable. Gaussian filtering is used as a smoothing technique [5], which in one dimension is given by:

$$X(f_S) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{f_S^2}{2\sigma^2}}, \tag{27}$$

where $\sigma$ is the standard deviation.

Following the above signal processing techniques, $\chi_{\text{in}}(t)$ and $\chi_{\text{out}}(t)$ are being processed and the corresponding algorithm flowchart is shown in Fig. 1. Data processing beat notes $\delta\nu_{in}(t)$ and $\delta\nu_{out}(t)$ (Eq. 15 and 16 at a sampling rate

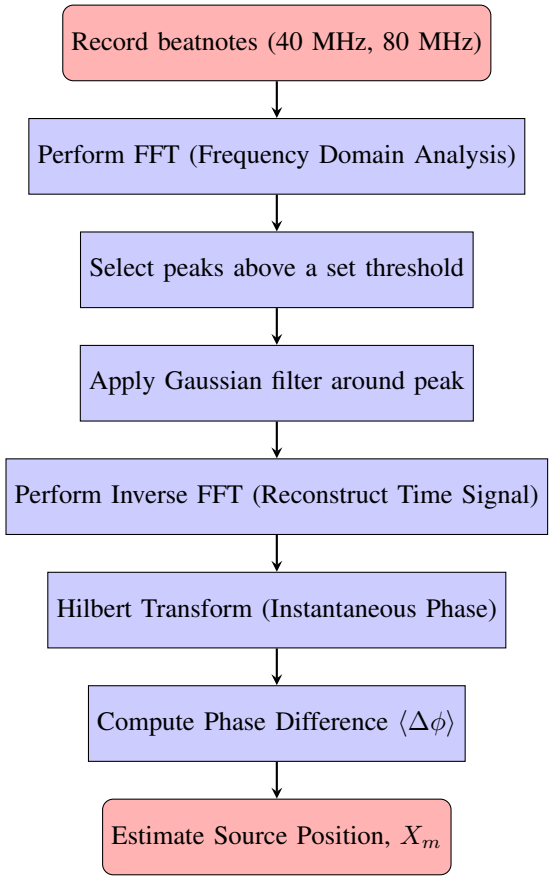



Fig. 1. Flow chart showing the data processing algorithm we used in out setup.

of 1000 Hz (Fig. 3(a) in the manuscript) followed by FFT (Eq. 23 for both the time series to obtain the frequency domain information (Fig. 3(b) in the manuscript). We then identify all the peaks above a certain excursion threshold, which is set to a value when number of peaks saturates. Then, we apply a Gaussian filter of required full width at half maximum (FWHM $\approx 2.355\sigma$) around the identical peaks from the post FFT data of $\delta\nu_{in}(t)$ and $\delta\nu_{out}(t)$. Then, we perform an IFT and get the real part and reconstruct the acoustic source such as excursion and frequency modulation of the disturbance over time. This assumes the acoustic source is sinusoidal in nature, $S(f_{S,i}, t) = S_\circ \sin(2\pi f_{S,i} t)$. Finally, a Hilbert transform of the IFT data provides instantaneous phase information, and their difference between the two time varying signals as given in Eq. 18 and 19, i.e. $\phi_{in,i} = [2\pi f_{S,i}(t - t_{L,i} - t_{R,i}) - \phi_{S,i}]$ and $\phi_{out,i} = [2\pi f_{S,i}(t - t_{L,i}) - \phi_{S,i}]$ gives,

$$\Delta\phi_i = \phi_{out,i} - \phi_{in,i} = 2\pi f_{S,i} t_{R,i}.$$

We take an average value of the phase difference $\langle\Delta\phi_i\rangle$ over a sample size in $2^n$-scale, where $n$ varies from 10 to 24, and use it to find the position of the source interacting with the fibre ($P$) as,

$$X_{m,i} = c_f \cdot t_{R,i} = \frac{c}{n_{\text{eff}}} \cdot \frac{\langle\Delta\phi\rangle}{2\pi f_{S,i}},$$

where $c$ is the speed of light in vacuum. For statistical understanding we repeat the process over several time samples corresponding $n$ = 10 to 24, of equal length to get the mean position $X_m$ and the standard deviation gives the inaccuracy $\Delta X_m$ in determining $X_m$. Also from the frequency spectrum after FFT analysis, we get the excursion $\chi_{\circ,i}$ produced by the $i^{\text{th}}$ acoustic source at a frequency $f_{S,i}$, and then using Eq. 11, we obtain displacement,

$$\mathcal{D}_i = \frac{\lambda_L}{2\pi n_{\text{eff}} f_{S,i}} \chi_{\circ,i}. \tag{28}$$

Further, following the concepts of simple harmonic motion, the peak-particle velocity (PPV) resulting from $S(f_{S,i}, t)$ is:

$$\nu_i = 2\pi f_{S,i} \mathcal{D}_i \tag{29}$$

$$= \frac{\lambda_L}{n_{\text{eff}}} \chi_{\circ,i}, \tag{30}$$

and the corresponding strain,

$$\varepsilon_i = \frac{\mathcal{D}_i}{L_{\circ,i}}, \tag{31}$$

where $L_{\circ,i}$ is the length of the fibre that is affected by $S(f_{S,i}, t)$. For example, while using the marker frequencies there a 2 m fibre is wrapped on the speaker, therefore $L_{\circ,i} =$ 2 m for them.

All data processing is performed using Python with the help of the following libraries: NumPy [6], SciPy [7], Pandas [8], and Matplotlib [9].

### C. Impact of a Super-Coherent Laser

An Ultra-stable and narrow linewidth, $\Delta\nu < 1$ Hz laser have low intrinsic phase noise and long coherence length $L_C = c/\pi\Delta\nu \simeq 10^8$m for $\Delta\nu = 1$ Hz, which are essential requirements to enhance the sensitivity of the interferometric sensing either homodyne base or heterodyne base detections [10], [11]. Any environmental noises perturbing the phase of such "super-coherent" photons propagating through the distributed fibre, therefore, adds phase noise, which in the SL-DAS scheme is being detected by the heterodyne technique and relevant information are extracted thereafter. The coherence length for a laser having $\Delta\nu = 1$ Hz is significantly longer than the present range of any optical fibre based DAS. This immense coherence length ensures phase correlation between the reference and the probe that is propagating through a sensing arm in a self-heterodyne scheme, enabling highly sensitive detection of the perturbations over a long distributed fibre. Other than narrow linewidth, low and linear drift of the laser's center frequency $\nu_L$ over time also plays a crucial role. Frequency drift $\Delta\nu_{\text{drift}}(t)$ of the laser resulting to a phase change over a period $t$ is given by;

$$\Delta\Phi_{\text{drift}}(t) = 2\pi \int_0^t \Delta\nu_{\text{drift}}(t')dt' = 0.21t \,\text{rad}, \tag{32}$$

which in specific to the super-coherent laser used in our system having $\Delta\nu_{\text{drift}} = 33\,\text{mHz/s}$. For an integration time of 1 ms as uesd in our experiment the laser acquires a phase change of 0.21mrad only. The interferometric detection is employed in SL-DAS where the light upon transmitting through the sensing fibre beats against a local reference derived from the same laser therefore, $\Delta\Phi_{\text{drift}}(t)$ appears as dc offset, only when $\Delta\nu_{\text{drift}}(t)$ is linear with time. This renders the SL-DAS immune to slow linear frequency drifts, enabling accurate phase demodulation and the detection of acoustic events. Therefore, the combination of narrow linewidth and low linear drift CW laser not only extends the sensing length but also guarantees phase stability of the probe laser, essential for robust interferometric sensing of acoustic signals. To support the importance of a super-coherent laser in SL-DAS, we numerically simulate the mean position $X_m$ on the fibre where an acoustic disturbance $S(S_0, f_S, x_0, y_0, z_0, t)$ interacts with the sensing fibre upon propagating via the elastic medium for varying laser properties e.g. stability and linewidth ($\Delta\nu$) as shown in Fig. 2(a) in the manuscript. The obtained results of determining $X_m$ with a certain accuracy $\Delta X_m$ for these laser properties are shown in Fig. 2(b), and the same is also compared with the experimental results.

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